

After marking the above program, and using *C-c C-*, the code becomes:

```
area           :: Int -> Int -> Int
area breadth height = breadth * height
```

Query

To know the Haskell-mode version, use *M-x* *haskell-version*. As an example:

Using *Haskell-mode* version *v2.8.0*,

C-c C-i on a symbol will prompt for getting information about the symbol. For example, ‘Show info of (default Int):’ lists the following:

```
data Int = GHC.Types.I# GHC.Prim.Int# -- Defined in GHC.Types
instance Bounded Int -- Defined in GHC.Enum
instance Enum Int -- Defined in GHC.Enum
instance Eq Int -- Defined in GHC.Base
instance Integral Int -- Defined in GHC.Real
instance Num Int -- Defined in GHC.Num
instance Ord Int -- Defined in GHC.Base
instance Read Int -- Defined in GHC.Read
instance Real Int -- Defined in GHC.Real
instance Show Int -- Defined in GHC.Show
```

C-c C-t will obtain the type of the symbol with the prompt ‘Show type of (default pi):’. For example:

```
pi :: Floating a => a
```

C-c TAB on a symbol returns its definition at the interpreter prompt, as follows:

```
*Main> :info sphereVolume
sphereVolume :: Fractional a => a -> a
-- Defined at /home/guest/Sphere.hs:1:1-12
```

To find Haddock information for a symbol, you can use *C-c C-d*. Searching for ‘Float’, for example, opens up the following file: <http://usr/share/doc/ghc/html/libraries/ghcprim-0.2.0.0/GHC-Types.html> on Fedora.

To use the Hayoo search engine, you can use *M-x* *haskell-hayoo*. It will prompt with:

Hayoo query:

The query responses are shown in a browser. Similarly, the Hoogle engine can be queried using *M-x* *haskell-hoogle*. If you searched for ‘show’, it will open the URL <http://www.haskell.org/hoogle/?q=show> with the search results.

Files ending with *.lhs* are literate Haskell programs. You can use the Richard Bird style to separate text and code as follows:

Insert blank line before the code

```
> quicksort :: Ord a => [a] -> [a]
> quicksort [] = []
> quicksort (p:xs) = (quicksort lesser) ++ [p] ++ (quicksort greater)
>   where
>     lesser = filter (< p) xs
>     greater = filter (>= p) xs
```

Insert blank line after the code

The modeline will indicate that you are in the ‘(Literate Haskell/bird)’ minor mode. The *hasktags* package needs to be installed to help generate TAGS file for source files. For example:

```
$ hasktags Test.hs
```

It will create both tags and TAGS files. You can use *M-* in the Haskell buffer to search for a tag.

Checks

HLint is a tool that provides suggestions to improve Haskell programs. *C-c C-v* helps to run *hlint* on a buffer. Make sure you have the tool installed on your system before using it. For example, running *C-c C-v* on the above literate quicksort Haskell program suggests:

```
*- mode: compilation; default-directory: "~/ -*-
Compilation started at Thu Jun 6 21:31:54
hlint QuickSort.lhs
QuickSort.lhs:6:22: Warning: Redundant bracket
Found:
(quicksort lesser) ++ [p] ++ (quicksort greater)
Why not:
    quicksort lesser ++ [p] ++ (quicksort greater)
QuickSort.lhs:6:44: Warning: Redundant bracket
Found:
[p] ++ (quicksort greater)
Why not:
[p] ++ quicksort greater
```



References

- [1] Haskell. <http://www.haskell.org/haskellwiki/Haskell>
- [2] GNU Emacs. <http://www.gnu.org/software/emacs/>
- [3] Haskell mode 2.8. <http://projects.haskell.org/haskellmode-emacs>
- [4] Hasktag. <http://hackage.haskell.org/package/hasktags>
- [5] HLint. <http://community.haskell.org/~ndm/hlint/>

By: Shakthi Kannan

The author is a free software developer at the Fedora project, and also a blogger. He co-maintains the Fedora Electronic Lab project.